

Noun Rank Benchmarking System v5.0

AI LLM Accuracy Validation for Long Document Summarization

GovReport-Calibrated Validation Algorithm

1. Overview

This document presents a benchmark-calibrated algorithm for validating AI-generated summaries using noun-based analysis. The system uses the GovReport dataset to establish baseline scores from expert-written summaries, then measures LLaMA 70B accuracy against these calibrated baselines.

1.1 Problem Statement

How to objectively measure the accuracy of AI LLM systems for Long Document Summarization using calibrated benchmarks and expert-validated ground truth.

1.2 Key Innovation

This version introduces a two-stage calibration approach:

- Calibration Stage:** Establish baseline scores using expert-written summaries from GovReport
- Evaluation Stage:** Measure LLaMA 70B accuracy as a percentage of expert baseline

This approach replaces the previous dual-model comparison (LLaMA vs Mistral) with a more rigorous expert-calibrated methodology.

1.3 System Specifications

Parameter	Value
Target Model	LLaMA 70B (INT8 quantization)
Validation Method	Noun Rank Algorithm (calibrated)
Benchmark Dataset	GovReport (19,466 documents)
Ground Truth	Expert-written summaries
Input	1–50-page documents
Output	1–5-page document summary + accuracy score

1.4 What Changed from v4

Component	v4 (Previous)	v5 (Current)
Validation Model	Mistral 7B	Not needed (removed)
Comparison Method	LLaMA vs Mistral scores	LLaMA vs Expert baseline
Ground Truth	None (relative comparison)	Expert summaries
ROUGE Metrics	Used for validation	Not needed (Noun Rank superior)
Trust Score	Score_L - Score_M	Score_L / Baseline

Infrastructure	Groq API	Vast.ai (recommended)
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2. Calibration Methodology

The system operates in two distinct stages to ensure rigorous validation against expert ground truth.

2.1 Stage 1: Baseline Calibration (One-Time Setup)

This stage establishes what a "perfect" summary scores by running Noun Rank on expert-written summaries from GovReport.

Process:

1. Load GovReport dataset (19,466 document-summary pairs)
2. For each document, run Noun Rank algorithm comparing source → expert summary
3. Store baseline score for each document in Reference Score Database
4. Calculate dataset statistics (mean, std, percentiles)

Output:

Document ID	Baseline Score
doc_0001	0.91
doc_0002	0.88
doc_0003	0.94
...	...
doc_19466	0.87

2.2 Stage 2: LLaMA Evaluation (Ongoing)

This stage measures LLaMA 70B accuracy against the calibrated baselines.

Process:

1. Load source document from GovReport
2. Generate summary using LLaMA 70B
3. Run Noun Rank algorithm comparing source → LLaMA summary
4. Retrieve baseline score for this document
5. Calculate accuracy: $\text{LLaMA Score} \div \text{Baseline Score} \times 100\%$

Accuracy Formula:

$$\text{Accuracy} = (\text{LLaMA Noun Rank Score} \div \text{Expert Baseline Score}) \times 100\%$$

Interpretation:

An accuracy of 92% means "LLaMA's summary is 92% as good as a human expert's summary" — a meaningful, objective measure.

2.3 Stage 3: Comparative Analysis

Aggregate results across the test set to generate comprehensive accuracy reports.

Document	Baseline	LLaMA Score	Accuracy	Status
doc_0001	0.91	0.87	95.6%	✓ Pass

doc_0002	0.88	0.82	93.2%	✓ Pass
doc_0003	0.94	0.71	75.5%	⚠ Review
Average	0.90	0.83	92.2%	-

3. Scoring Components

The Noun Rank algorithm uses five weighted components to calculate a comprehensive score.

Component	Symbol	Weight	Purpose
Hallucination Score	H	0.25	Penalize fabricated content not in source
Coverage Score	C	0.25	Reward inclusion of key facts from source
Frequency Score	F	0.20	Weight nouns by source importance
Entity Score	E	0.20	Validate named entities strictly
Semantic Score	S	0.10	Overall meaning alignment

Final Score Formula:

$$\text{Score} = (1 - H) \times 0.25 + C \times 0.25 + F \times 0.20 + E \times 0.20 + S \times 0.10$$

3.1 Why (1 - H) in the Formula?

The Hallucination Score (H) is a penalty metric where higher values indicate more hallucinations. Using (1 - H) inverts the direction so all components align — higher always means better.

H Value	Meaning	(1 - H)	Contribution
0.00	Perfect	1.00	Maximum
0.10	Excellent	0.90	High
0.25	Acceptable	0.75	Medium
0.35	Invalid	0.65	Low

4. Algorithm Steps

Step 1: Extraction & Normalization

Extract from source document D1:

- Common Nouns (lemmatized) → Set: D_nouns
- Proper Nouns → Set: D_proper
- Named Entities → Set: D_entities (ORG, PERSON, GPE, DATE, MONEY)
- Numbers/Dates → Set: D_numbers
- Noun Frequencies → Dict: D_freq {noun: count}

Key Nouns: $D_key = \{n : n \in D_nouns \text{ AND } D_freq[n] \geq 2\}$

Explanation: Key nouns are nouns that appear two or more times in the source document. These frequently mentioned nouns represent the core topics that the summary must cover.

Step 2: Hallucination Score (H)

Calculate soft threshold instead of binary validation:

1. $H_common = |S_nouns - D_nouns| / |S_nouns|$
2. $H_proper = |S_proper - D_proper| / |S_proper|$
3. $H_entity = |S_entities - D_entities| / |S_entities|$

Weighted Formula: $H = (H_common \times 0.3) + (H_proper \times 0.3) + (H_entity \times 0.4)$

H Value	Interpretation
$H = 0.00$	Perfect (no hallucinations)
$H < 0.10$	Excellent
$H < 0.20$	Acceptable
$H \geq 0.20$	Suspicious
$H \geq 0.35$	Likely invalid

Synonym Tolerance: Before marking a noun as unmatched, check lemmatized form, synonym set (WordNet), and embedding similarity > 0.85.

Step 3: Coverage Score (C)

Measures whether the summary includes important facts from the source:

$$C_noun = |S_nouns \cap D_key| / |D_key|$$

$$C_entity = |S_entities \cap D_entities| / |D_entities|$$

$$C_number = |S_numbers \cap D_numbers| / |D_numbers|$$

Weighted Formula: $C = (C_noun \times 0.4) + (C_entity \times 0.4) + (C_number \times 0.2)$

Step 4: Frequency Score (F)

Uses log-normalized frequency to prevent extreme values from dominating:

For each noun n in S_nouns: if $n \in D_nouns$, $F_sum += \log(1 + D_freq[n])$

Raw frequency score: $F_raw = F_sum / |S_nouns|$

Normalize: $F = F_raw / \text{max_possible_F_raw}$

Step 5: Entity Score (E)

Named entities are validated with strict matching:

Entity Type	Weight	Notes
PERSON	1.0	Names must be exact
ORG	1.0	Organizations
GPE	1.0	Locations
DATE	0.8	Dates
MONEY	0.8	Monetary values
PERCENT	0.8	Percentages

Step 6: Semantic Score (S)

Overall meaning alignment using sentence embeddings:

$\text{emb_D1} = \text{embed}(\text{D1})$ — Source document embedding

$\text{emb_S} = \text{embed}(\text{S})$ — Summary embedding

$S = \text{cosine_similarity}(\text{emb_D1}, \text{emb_S})$

Typical range: 0.70 - 0.95 for good summaries

Step 7: Final Score & Accuracy Calculation

Calculate Noun Rank score and compare against baseline:

1. Calculate Score: $\text{Score} = (1 - H) \times 0.25 + C \times 0.25 + F \times 0.20 + E \times 0.20 + S \times 0.10$
2. Retrieve Baseline: $\text{baseline} = \text{reference_scores}[\text{document_id}]$
3. Calculate Accuracy: $\text{accuracy} = (\text{score} / \text{baseline}) \times 100\%$

5. Accuracy Interpretation

The accuracy score represents how well LLaMA's summary compares to an expert-written summary for the same document.

Accuracy	Level	Interpretation
95-100%	Excellent	Near expert quality, production ready
85-94%	Good	High quality, acceptable for most use cases
70-84%	Acceptable	Moderate quality, may need review
50-69%	Low	Significant issues, requires manual review
<50%	Reject	Unacceptable, do not use

5.1 Decision Matrix

Condition	Action
H $\geq$ 0.35	REJECT — Hallucination threshold exceeded
Accuracy $\geq$ 95%	ACCEPT — High confidence, no review needed
Accuracy $\geq$ 85%	ACCEPT — Good quality, spot check recommended
Accuracy $\geq$ 70%	REVIEW — Manual verification recommended
Accuracy < 70%	REJECT — Below acceptable threshold

6. GovReport Benchmark Dataset

GovReport serves as both the calibration source and evaluation benchmark for the Noun Rank system.

6.1 Dataset Overview

Feature	Value
Total Documents	19,466 document-summary pairs
Average Document Length	~9,400 words
Average Summary Length	~550 words
Summary Authors	Government experts (CRS & GAO)
Train/Val/Test Split	17,517 / 974 / 975
Access	Hugging Face: ccdv/govreport-summarization

6.2 Why GovReport is Ideal

- **Expert-Written Summaries** — Ground truth written by the same experts who authored the reports
- **Long Documents** — Average 9,400 words matches target use case of 1–50-page documents
- **Factual Content** — Government reports contain verifiable facts, entities, and data points
- **Formal Structure** — Professional documents with clear sections for reliable NLP processing
- **Free & Accessible** — Available on Hugging Face with no licensing restrictions

6.3 Dataset Access Code

```
# Install and load GovReport dataset
pip install datasets
from datasets import load_dataset
govreport = load_dataset("ccdv/govreport-summarization")
```

7. System Architecture

This section defines the recommended technology stack and infrastructure for implementing the Noun Rank Benchmarking System.

7.1 Building Blocks

Component	Technology
Target Model	LLaMA 70B (INT8 quantization)
Validation Method	Noun Rank Algorithm
LLM Inference	Vast.ai (recommended) — see Section 7.2
Benchmark Dataset	GovReport (19,466 documents)
NLP Library	spaCy (noun extraction, NER, lemmatization)
Embeddings	sentence-transformers
Programming Language	Python
Orchestration	Raspberry Pi / Local Machine

7.2 Infrastructure Selection: Vast.ai (Recommended)

★ **RECOMMENDED:** Vast.ai is selected as the primary infrastructure provider for better control, flexibility, and high-volume processing capabilities.

Vast.ai vs Groq Comparison

Factor	Vast.ai ★	Groq (Reference)
Speed	~50-150 tokens/sec	~500 tokens/sec
Model Control	Full control (custom models)	Pre-selected models only
Fine-tuning	Available	Not available
CUDA Access	Full GPU access	Not available
Cost Model	Hourly rental	Per-token pricing
High Volume	Cost-effective at scale	Expensive at scale
Setup Time	30+ minutes	5 minutes
Maintenance	Self-managed	Fully managed

7.3 Why Vast.ai for This Project

- **Full System Control** — Direct access to GPU, CUDA, and model weights for debugging and optimization
- **Benchmark Processing** — Processing 19,466 GovReport documents is cost-effective with hourly rental
- **Custom Model Support** — Ability to run fine-tuned or custom quantized models
- **Learning Opportunity** — Hands-on experience with GPU infrastructure and model deployment
- **Batch Processing** — Efficient for calibration stage (one-time processing of entire dataset)

7.4 When to Use Groq Instead (Reference)

Groq remains a viable option for specific scenarios:

- Quick prototyping and development (free tier available)
- Low-volume production (<100 documents/day)
- When speed is critical and cost is secondary
- Zero-maintenance requirement

7.5 Recommended Vast.ai Configuration

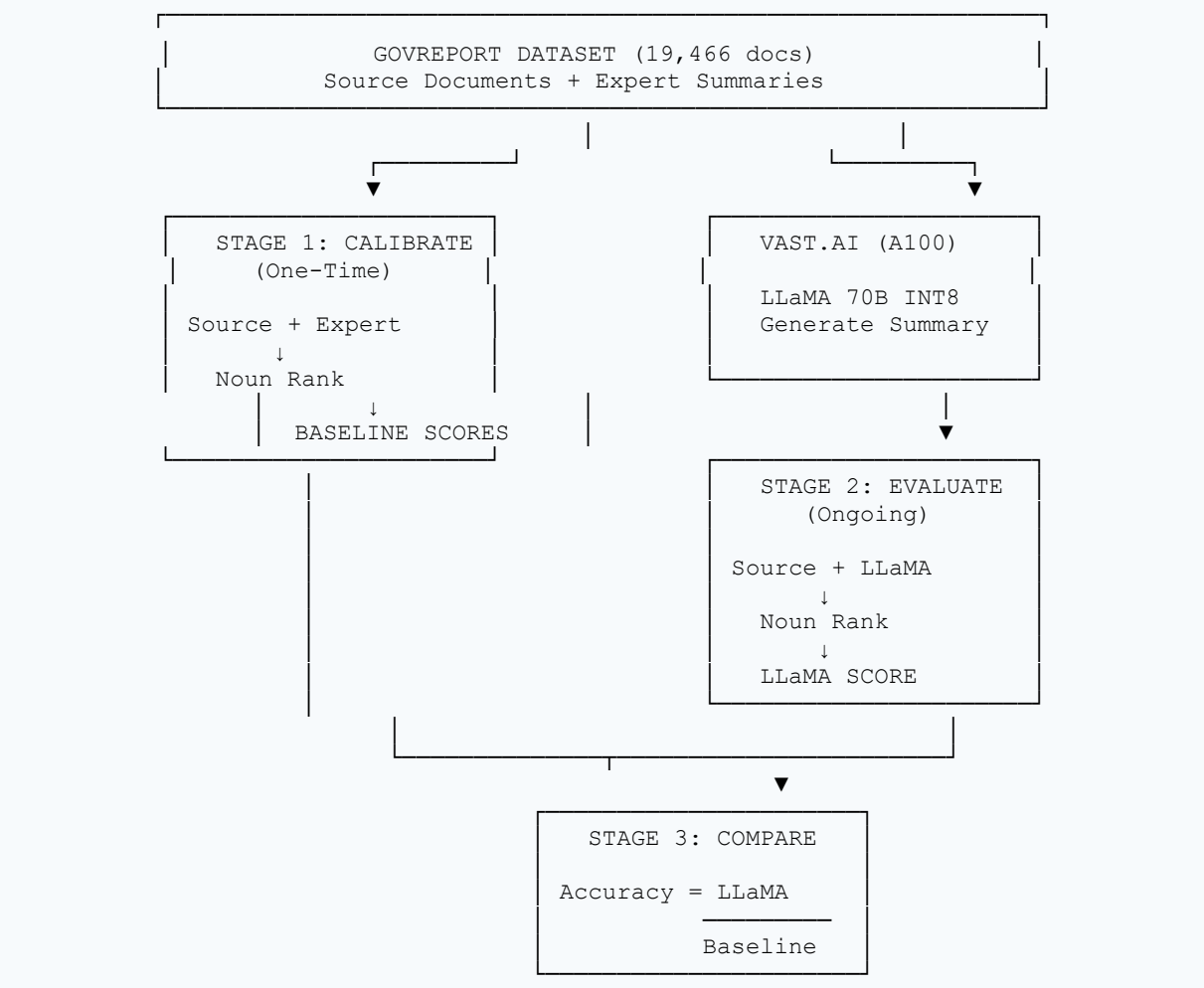
Component	Specification
GPU	NVIDIA A100 80GB
VRAM Required	~70GB for LLaMA 70B INT8
Disk Space	100GB+ (model + dataset)
Docker Image	pytorch/pytorch:latest or vllm
Estimated Cost	~\$.7 - 1/hour

7.6 Cost Estimation

Task	Estimated Cost (Vast.ai)
Calibration (19,466 docs, one-time)	~\$50-100 (20-40 hours)
Test Set Evaluation (975 docs)	~\$5-10 (2-4 hours)
Production (100 docs/day)	~\$3-5/day
Development/Testing	~\$2-5/session

8. Architecture Diagram

The system architecture separates calibration (one-time) from evaluation (ongoing).



9. Implementation Requirements

9.1 Required Libraries

- **spaCy** — Noun extraction, NER, lemmatization
- **NLTK WordNet** — Synonym matching
- **sentence-transformers** — Embedding similarity
- **datasets** — Hugging Face dataset loading
- **vllm or transformers** — LLM inference on Vast.ai

9.2 Setup Commands

```
# Install dependencies
pip install spacy nltk sentence-transformers datasets vllm

# Download spaCy model
python -m spacy download en_core_web_sm

# Download NLTK WordNet
python -c "import nltk; nltk.download('wordnet')"
```

9.3 Complexity Estimate

Component	Difficulty
Noun extraction	Easy
NER	Easy
Lemmatization	Easy
Synonym matching	Medium
Embedding similarity	Medium
Vast.ai setup	Medium
Full system	Medium (~600 lines of code)

10. Success Criteria

The following metrics define success for the Noun Rank Benchmarking System.

Metric	Target
Hallucination Detection Accuracy	$\geq 85\%$ (vs manual review)
Average LLaMA Accuracy	$\geq 85\%$ of expert baseline
Baseline Score Consistency	$\sigma < 0.10$ across dataset
Processing Speed (validation)	≤ 30 seconds per document
Processing Speed (LLaMA)	≤ 60 seconds per document

11. Thresholds Summary

Metric	Excellent	Good	Acceptable	Reject
Hallucination (H)	< 0.05	< 0.15	< 0.25	>= 0.35
Coverage (C)	> 0.80	> 0.65	> 0.50	< 0.40
Frequency (F)	> 0.80	> 0.60	> 0.40	< 0.30
Entity (E)	> 0.90	> 0.75	> 0.60	< 0.50
Semantic (S)	> 0.90	> 0.80	> 0.70	< 0.60
Accuracy	> 95%	> 85%	> 70%	< 50%

12. Conclusion

Version 5.0 of the Noun Rank system introduces a rigorous, benchmark-calibrated methodology for validating AI-generated summaries.

Key Innovations:

- **Expert-Calibrated Baselines** — Using GovReport's expert summaries as ground truth
- **Absolute Accuracy Measurement** — LLaMA accuracy as percentage of expert quality
- **Simplified Architecture** — Removed Mistral validation model (replaced by expert baselines)
- **ROUGE Replacement** — Noun Rank provides superior validation over traditional ROUGE metrics
- **Vast.ai Infrastructure** — Full control for benchmark processing and model deployment

This approach transforms the Noun Rank system from a relative comparison tool into a calibrated benchmarking system that measures LLM accuracy against expert human performance.

The result: When the system reports "92% accuracy," it means LLaMA's summary is 92% as good as an expert-written summary — a meaningful, objective, and actionable metric.